

Did England Deserve to Win? A Data Dashboard

Analysis of UEFA Women's Euro 2025

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Live dashboard: <https://jbuergler.github.io/football-analytics/>

Repository: <https://github.com/jbuergler/football-analytics>

1 Introduction and Research Question

The UEFA Women's European Championship (Euro) is one of the most important tournaments in women's football. The 2025 edition, hosted in Switzerland from 2nd to 27th July 2025, came at a moment when the sport continues to grow in visibility, commercial value and public attention. The tournament has achieved record television audiences in the United Kingdom (cite: womens sport trust 2026). Sixteen nations competed across the group stage and knockout rounds, producing 106 goals across 31 matches and setting multiple tournament records (cite: uefa records article).

Research on women's football shows that this growth is part of a broader professionalisation, but that major gaps between the women's and men's game remain in areas such as commercial investment, media coverage, and resources (cite: tjonndal 2024 journal). This makes the tournament interesting not just because of who won, but about whether performance data

supports the results. This is something that matters to fans, analysts, journalists, coaches, and anyone interested in how women's football is judged and valued.

The tournament ended with the final match at St. Jakob on the 27th of July 2025, where England faced Spain. England entered the tournament as the defending title holders, whereas Spain were the reigning World Cup holders. Spain's Aitana Bonmatí was named Player of the Tournament, while Esther González finished as the tournament's top scorer with four goals (cite: uefa top scorer article, uefa team of tournament article). Following a 1-1 draw after extra time, England won 3-1 on penalties to retain their European title (cite bbc sport final article).

The result, however, raised an immediate analytical question: did the scoreline reflect the balance of the play throughout the tournament and in the final itself? This project uses StatsBomb open event data to directly compare England and Spain's performances across the tournament and in the final itself, through metrics including expected goals (xG), pressing intensity, and ball progression. The central research question is:

Did England deserve to win the Women's Euro 2025?

This question matters because women's football is undergoing clear professionalisation, yet there are still major gaps with the men's game regarding visibility, financial conditions, and commercial value (cite: tjondall). Recent research also shows that professional women's football has evolved technically and tactically, with event data revealing longer possession sequences, fewer long-distance shots, and more difficult, valuable passes (cite bransen and davis). This makes event data a useful basis for assessing whether the result reflected the balance of play. For that reason, this project evaluates whether England's title win in 2025 was deserved on performance as well as on the scoreboard.

2 Data Collection and Preparation

2.1 Source and Access

For this project, event data for the UEFA Women's EURO 2025 was sourced via the StatsBombR package in R (cite: statsbomb open data). Recently, StatsBomb was incorporated into Hudl's industry-leading ecosystem of integrated video and analytic solutions (source hudl + statsbomb 2024). They are an established provider of football event data, recording over 3,400 events per match across more than 190 competitions and thus providing a wide range of data free of charge (hudl statsbomb source)

2.2 Dataset Structure and Granularity

The raw data is structured at the event level, meaning that each row represents a single action during a match. This includes actions such as passes, carries, shots, pressures, and ball receipts. Each event has information on the player, team, match, time, pitch location (x, y coordinates), and action-specific attributes such as shot outcome and xG value for shots.

The full dataset covers 31 matches across the group stage and knockout rounds, including 875 shots after cleaning. Key variables used in this analysis include:

- **location_x, location_y**: pitch coordinates of the action
- **shot.statsbomb.xg**: expected goals value for each shot
- **shot.outcome.name**: outcome of the shot (goal, saved, off target, etc.)

- **type.name:** event type (shot, pass, carry, pressure, etc.)
- **team.name:** the team performing the action
- **player.name:** the player performing the action
- **minute, period:** time context of the action

2.3 Cleaning and Preparation

Several cleaning steps were applied to prepare the data for the analysis:

Period filter: StatsBomb records penalty shootouts as period 5. As penalty shootouts are based on individual shots rather than open play performance, all period 5 events were excluded from the analysis. This affected a total of three matches: the Sweden vs England semi-final, the France vs Germany quarter final, and the England vs Spain final.

Own goals: Own goals were excluded from the shot and xG analysis as they do not reflect the performance of the attacking team. This means goals from the analytical shot count (875 shots) differs slightly from the total goals shown in the dashboard (103 vs 106).

Team name standardisation: Team names were cleaned and standardised to ensure consistency across all tables and visualisations. Prefixes and Suffixes like “Women’s” or “WNT” (for Finland) were removed.

Pre-aggregation: Summary tables for the different metrics analysed in this project were computed in advance and saved as .rds files. They are loaded directly by the Shiny dashboard. This approach improves loading speed and separates the data preparation from the app itself.

2.4 Coverage and Caveats

As shown in Table 1, the dataset covers all 31 matches of the UEFA Women’s Euro 2025 tournament from the group stage through to the final. As coverage is subject to StatsBomb’s data collection methodology, any recording errors cannot be ruled out.

This project analysed on-ball actions only. Off-ball movements and physical metrics such as distance covered are not included. This means that some aspects of performance, especially defensive organisation away from the ball, were not covered in this analysis. StatsBomb’s freeze frame data, which records the location of all players involved in the event, including goalkeeper and defender positions at the moment of each shot (hudl statsbomb 2021 freeze frames) was not used in this analysis. However, it informed the pre-computed xG values available in the dataset, which makes them more contextually accurate. This directly represents a potential aspect, that future work could incorporate in the analysis. Across the full dataset, a total of 105,525 individual events were recorded.

StatsBomb records concrete pressing events when a player applies pressure to an opponent in possession, instead of computing pressure through combining multiple defensive actions (cite: dorrington pressure data in football 2025). This is an advantage, as it allows pressing intensity to be evaluated directly from recorded events.

Table 1: Table 1: Dataset Summary - UEFA Women's Euro 2025

Item	Detail
Source	StatsBomb open event data
Access method	StatsBomb R package
Competition	UEFA Women's Euro 2025 (ID: 53, Season: 315)
Matches	31
Shots	875
Granularity	Event level (one row per action), a total of 105,525 events
Period 5 excluded	Yes (penalty shootouts)
Own goals excluded	Yes (from xG analysis)

3 Methodology

3.1 From Raw Data to Dashboard

This project follows a structured pipeline of five scripts, moving from raw data collection through to the interactive dashboard. Raw event data was pulled from StatsBomb in script 01_data.R, explored in 02_explore.R, cleaned in 03_clean.R, aggregated into summary tables in 04_analyse.R, and visualised in 05_visualise.R. As a first step, only the planned static visuals were created, before moving them as interactive elements in app/app.R. After the app successfully ran locally on R, it was then deployed as the dashboard itself via Shinylive to GitHub. This ensures the analysis is reproducible and the dashboard only uses smaller, cleaned data tables, that are easier to load through Shiny.

3.2 Metrics

All metrics were computed from the cleaned event data and aggregated into summary tables. The following metrics were used across the dashboard:

Expected goals difference (xG diff): For each match, xG difference was calculated as a team's total xG minus their opponent's total xG. Averaging this across all matches gives a tournament-wide measure to see whether a team consistently created better chances than they conceded. This is the primary metric for the Tournament Overview tab of the Dashboard.

High press rate: The high press rate was defined as the proportion of all pressure events happening in the opponent's half of the pitch (`location_x > 60` on the StatsBomb coordinate system, where the pitch runs from 0 to 120). StatsBomb records a pressure event each time a player actively applies pressure to an opponent in possession, which is a direct measure of pressing behaviour.

Counter-press rate: The counter-press rate was defined as the proportion of events that StatsBomb flagged as counter pressing. This refers to pressures applied within a short window of losing the ball. Missing values were recoded to FALSE so the full pressure count could be used to compute the rate in %.

Pass completion rate: The proportion of attempted passes that successfully reached a teammate from all open play passes. Unknown outcomes and injury clearances from `pass.outcome.name` were excluded. It is worth noting that in the StatsBomb data, NA values in `pass.outcome.name` refer to completed passes (statsbomb pdf source).

Final third carry rate: This refers to the proportion of all carries ending beyond $x = 80$ on the StatsBomb coordinate system, which represents the final third of the pitch. This metric was chosen over the standard StatsBomb progressive carry definition (carries advancing the ball by at least 10 units). As it directly shows penetration into the attacking third instead of forward progress from any location, this was chosen as a more meaningful metric to analyse the attacking play.

Final third pass rate: The proportion of completed passes that ended beyond $x = 80$ to measure how frequently each team moved the ball into the attacking third through passing.

Shots faced per match: The average number of shots conceded per match, used as a measure for defensive solidity alongside average xG conceded.

3.3 Grouping and Filtering Decisions

All pressing metrics were split by stage type (group stage vs knockout rounds) to capture if there were any tactical shifts as the tournament progressed. The final match was isolated by selecting only the match ID of the final itself using `competition_stage == "Final"` and analysed separately across the tabs that focused on a detailed analysis of the events in the final between England and Spain. Actions of the best players were filtered to Bonmatí and Hemp specifically. They accumulated the most final third passes and carries for their teams.

3.4 Dashboard Design

The dashboard was built using R Shiny with the `bslib` package for the layout and theme. A consistent colour palette was applied for the entire dashboard based on the official hex colours of Spain (`#F1BF00`) and England (`#CE1124`). All other teams were highlighted in grey to ensure England and Spain can be immediately identified across every chart.

The dashboard was structured across a total of six tabs that guides the reader through the story of the tournament and ultimately answering the research question. First, Tournament Facts and Context was provided about the competition itself, as well as manager, formation, and high-performing player information for England and Spain. After that, the Tournament Overview provided context of the team's performance regarding their expected goals (scored and conceded). Moving on, the Journey Tab shows the development of the two finalists throughout the tournament across metrics like accumulated xG, match-by-match xG and pressing. Afterwards, the final was analysed in more detail across two tabs to analyse the shot and xG situation, as well as the Possession in separate windows. The final was split across two tabs deliberately, as otherwise the plots would be too visually cluttered and overwhelming. Finally, an analytical decision was made based on the analysed metrics to answer the research question. This ensures that the reader is guided from a broad to specific context. Each tab includes a navigation sidebar that can be collapsed if need be, to give some initial context of what each tab is trying to show. A preview of the Dashboard can be found in Appendix A.

The individual chart types were chosen based on the analytical question each visualisation attempted to address. A horizontal bar chart (Tab 1) ranks all 16 teams by xG dominance,

making the ranking immediately readable. A scatter plot (Tab 1) compares total xG against goals scored, showing which teams over- or underperformed in terms of goals. Cumulative line charts show how the xG and momentum changes across the tournament, while step charts show the xG accumulated shot by shot in the final. Pitch-based plots like shot maps (scatter plot) and player passmaps (arrow maps) use spatial data to show where on the pitch certain actions happened.

All plotly charts include hover tooltips providing exact values for each elements of the plots. Two plots (xG Difference horizontal bar chart and player passmaps), however, were deliberately kept as static plots using `renderPlot` instead of `plotly`. This was due to compatability constraints between the plots and the rendering mechanism of `plotly` in the Shinylive WebAssembly environment.

4 Findings

5 Discussion, Limitations, and Conclusion

List of Tools Used

Anara Technologies (2026) *Anara* (Pro Version). Available at: <https://anara.com>

- Assisted in understanding and summarising academic literature
- Facilitated the organisation of research findings

Anthropic (2026) *Claude* (Version claude-sonnet-4-6). Available at: <https://claude.ai>

- Assisted in debugging R code and Shiny app development
- Support with structuring report sections
- Proofreading and grammar check for cohesion and readability

Grammarly Inc. (2026) *Grammarly* (Free Version). Available at: <https://www.grammarly.com>

- Support with spelling and grammar checks, including punctuation and preposition use

QuillBot (2026) *Paraphrasing Tool* (Free Version). Available at: <https://quillbot.com/paraphrasing-tool>

- Assisting with paraphrasing sentences and sentence structures
- Finding synonyms

Zotero (2026) *Zotero* (MacOS Version). Available at: <https://www.zotero.org/>

- Assisting in the organisation of references and citations

Appendices

5.1 Appendix A: Dashboard Preview

Bibliography